### 🕒 1. What Are Dates in Python?

Python doesn't have a built-in data type for dates, but it provides the **datetime** module to work with dates and times easily.

```
import datetime
```

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### 🗓️ 2. Getting the Current Date and Time

```
import datetime


current = datetime.datetime.now()

print(current)
```

🧠 **Explanation:**

- datetime.now() gives the **current local date and time**.
- Example output:
- 2025-10-21 13:05:42.123456

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### 🗓️ 3. Getting Only the Date

```
import datetime


today = datetime.date.today()

print(today)
```

**Output:**

2025-10-21

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### 🧩 4. Creating a Specific Date

You can create a date object manually:

```
import datetime
```

```
my_birthday = datetime.date(1990, 12, 25)

print(my_birthday)
```

**Output:**

```
1990-12-25
```

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### ⏰ 5. Getting Individual Components

```
import datetime


now = datetime.datetime.now()


print("Year:", now.year)

print("Month:", now.month)

print("Day:", now.day)

print("Hour:", now.hour)

print("Minute:", now.minute)

print("Second:", now.second)
```

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### 🧮 6. Date Formatting (strftime())

Convert a date to a readable string format.

```
import datetime


now = datetime.datetime.now()


print(now.strftime("%A, %d %B %Y"))   # Tuesday, 21 October 2025

print(now.strftime("%m/%d/%Y"))       # 10/21/2025
```

```
print(now.strftime("%I:%M %p"))      # 01:15 PM
```

📘 **Common Format Codes**

| Code | Meaning | Example |
|---|---|---|
| %Y | Year (4-digit) | 2025 |
| %y | Year (2-digit) | 25 |
| %B | Month name | October |
| %b | Short month | Oct |
| %A | Weekday name | Tuesday |
| %a | Short weekday | Tue |
| %d | Day of month | 21 |
| %m | Month number | 10 |
| %H | Hour (24-hour) | 13 |
| %I | Hour (12-hour) | 01 |
| %M | Minutes | 15 |
| %S | Seconds | 42 |
| %p | AM/PM | PM |

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🔁 **7. Parsing String to Date (strptime())**

Convert a **string** into a **datetime object**.

```
from datetime import datetime

date_string = "21/10/2025 14:30"
date_obj = datetime.strptime(date_string, "%d/%m/%Y %H:%M")
```

```
print(date_obj)
```

**Output:**

```
2025-10-21 14:30:00
```

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### ➕➖ 8. Date Arithmetic (Add/Subtract Dates)

You can use timedelta to add or subtract days, weeks, or hours.

```
from datetime import datetime, timedelta

today = datetime.now()
next_week = today + timedelta(days=7)
yesterday = today - timedelta(days=1)

print("Today:", today)
print("Next week:", next_week)
print("Yesterday:", yesterday)
```

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### 📏 9. Difference Between Two Dates

```
from datetime import date

d1 = date(2025, 10, 21)
d2 = date(2024, 10, 21)

diff = d1 - d2
print(diff.days, "days")
```

**Output:**

```
365 days
```

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### ⏳ 10. Working with Time Only

```
from datetime import time


t = time(14, 30, 45)
print(t)
print("Hour:", t.hour)
```

**Output:**

```
14:30:45
Hour: 14
```

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### ✅ Summary:

| Task | Function |
|---|---|
| Get current date/time | datetime.now() |
| Get today's date | date.today() |
| Create specific date | date(YYYY, M, D) |
| Add/Subtract days | timedelta(days=...) |
| Format date to string | strftime() |
| Parse string to date | strptime() |